# GANDHI INSTITUTE OF ENGINEERING AND TECHOLOGY UNIVERSITY, ODISHA, GUNUPUR

## Department of CSE

## Minor Project-I Review Report-1

| PROJECT TITLE | | | Group Number | |
|---|---|---|---|---|
| **Team Leader Name and Roll No** | **Team Members Name and Roll No** | **Minor Project-I Status** | **Tentative Date of Submission of Project** | |
| **1.Biswajit Behera (23CSEAIML086)** | **2.Ayushman Abhisek Mallick (23CSEAIML101)**<br>**3.Swastik Panda (23CSEAIML124 )** | 2nd July 2025 to 17th Aug 2025 | 22nd to 28th October 2025 | |

# PROJECT SUMMARY: -

**The primary aim of this project is to develop an automated system capable of accurately classifying plant leaf images into their respective disease categories. By leveraging deep learning techniques, this solution intends to aid farmers, agricultural researchers, and plant pathologists in early detection and treatment of plant diseases, thereby improving crop yield and reducing economic losses.**

**This project uses a Convolutional Neural Network (CNN) built with TensorFlow and Keras, trained on the New Plant Diseases Dataset from Kaggle containing images of healthy and diseased leaves from multiple plant species. The dataset is fetched via the Kaggle API in Google Colab, with preprocessing steps like resizing, normalization, and augmentation to improve generalization. The custom CNN, optimized with Adam and categorical cross-entropy loss, is trained on Colab's GPU and evaluated using accuracy, precision, recall, and confusion matrix. The final model is saved in .h5 format and integrated into a Streamlit web app for real-time plant disease detection.**

# PROJECT OVERVIEW: -

| CATEGORY* | PRESENT STATUS OF PROJECT | PERCENTAGE OF WORK COMPLETED | PENDING WORK |
|---|---|---|---|
| Agriculture Machine Learning Model | Dataset preprocessing completed<br>Evaluation metrics generated<br>Model trained | 40% | Model fine-tuning for improved accuracy<br>Building and integrating Streamlit UI<br>Testing web app with sample images |

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***(Research Based/App Development (system/mobile/cloud/iot, etc.)/Web Application/Data Science Application)**

## Comments of the Supervisor: -

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**Mr. Tirupati Sahu**

**Signature of Supervisor**